

Case Study 3

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The **.Rmd** and **.html** (or **.pdf**) should be uploaded in TUWEL by the deadline. Refrain from using explanatory comments in the R code chunks but write them as text instead. Points will be deducted if the submitted file is not in a decent form.

Data

Load the data set you exported in the final Task of Case Study 2. Eliminate all observations with missing values in the income status variable.

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.2.0
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.3      ✓ tibble     3.2.1
## ✓ lubridate  1.9.5      ✓ tidyr      1.3.1
## ✓ purrr      1.0.4
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```
library(ggplot2)
```

```
library(forcats)
```

```
ds <- read.csv2('df_vars.csv', na.strings = '.', dec = '.')
ds <- drop_na(ds, income_class)
ds <- mutate(ds, income_class = factor(income_class, levels = c("H", "UM", "LM", "L")))
head(ds)
```

country <chr>	G... continent <chr><chr>	education_expenditures <dbl>	net_migration_rate <dbl>	youth_...
1 Afghanistan	AFG Asia	4.1	-0.1	
2 Albania	ALB Europe	3.6	-3.3	
3 Algeria	DZA Africa	NA	-0.9	
4 American Samoa	ASM Oceania	NA	-26.1	
5 Andorra	AND Europe	3.2	0.0	
6 Angola	AGO Africa	3.4	-0.2	

6 rows | 1-7 of 9 columns

As a reminder, the data set includes world data from 2020, focusing on:

- **Education Expenditure (% of GDP)**
- **Youth Unemployment Rate (15-24 years)**
- **Net Migration Rate** (difference between the number of people entering and leaving a country per 1,000 persons)

for most world entities in 2020. The data was downloaded from <https://www.cia.gov/the-world-factbook/about/archives/> (<https://www.cia.gov/the-world-factbook/about/archives/>). Additional information on continent, subcontinent/region and income status was appended to the dataset in Case Study 2.

Tasks:

a. Education expenditure in different income levels

Using **ggplot2**, create a density plot of the education expenditure grouped by income status. The densities for the different groups are superimposed in the same plot rather than in different plots. Ensure that you order the levels of the income status such that in the plots the legend is ordered from High (H) to Low (L).

- The color of the density lines is black.
- The area under the density curve should be colored differently among the income status levels.
- For the colors, choose a transparency level of 0.5 for better visibility.
- Position the legend at the top center of the plot and give it no title (hint: use `element_blank()`).
- Rename the x axis as "Education expenditure (% of GDP)"

Comment briefly on the plot.

```
ggplot(ds, aes(x = education_expenditures, fill = income_class, color = income_class)) +  
  geom_density(color = "black", alpha = 0.5) +  
  labs(x = "Education expenditure (% of GDP)", y = "Density", title = "Distribution of Education Expenditure by Income Status") +  
  theme(legend.position = "top",  
        legend.justification = "center",  
        legend.title = element_blank())  
)
```

```
## Warning: Removed 48 rows containing non-finite outside the scale range  
## (`stat_density()`).
```

Distribution of Education Expenditure by Income Status



Low-income countries tend to have the lowest education expenditure, with a very broad distribution concentrated at low values. Lower-middle and upper-middle income countries show similar central tendencies, though the LM group appears more spread out. High-income countries tend to spend the most, with a pronounced peak around 5% of GDP.

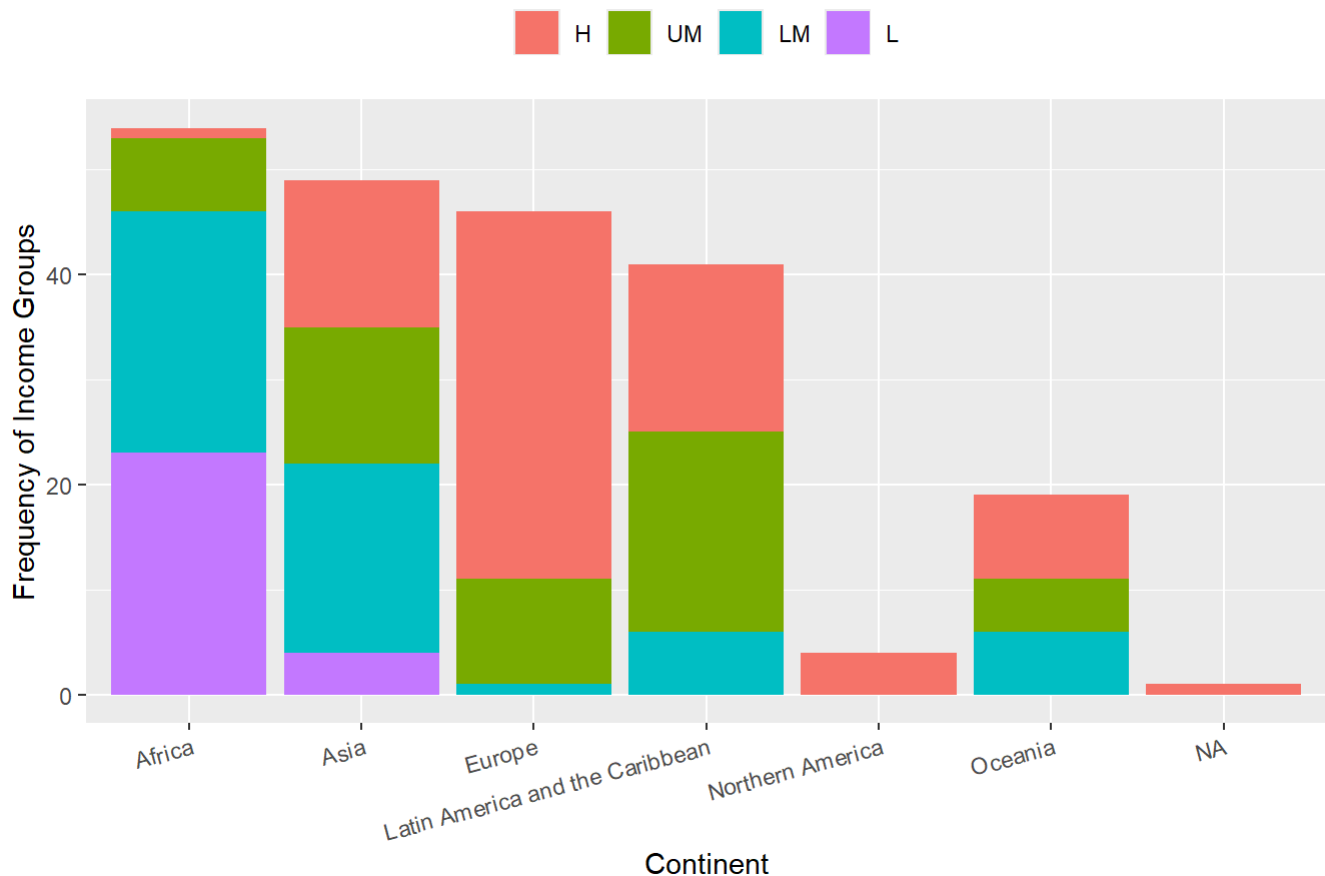
b. Income status in different continents

Investigate how the income status is distributed in the different continents.

- Using **ggplot2**, create a stacked barplot of absolute frequencies showing how the entities are split into continents and income status. Comment the plot.

```
ggplot(ds, aes(x = continent, fill = income_class)) +
  geom_bar(position = "stack") +
  labs(x = "Continent", y = "Frequency of Income Groups", title = "Income Status Distribution
by Continent (Absolute Frequencies)") +
  theme(legend.position = "top",
        legend.justification = "center",
        legend.title = element_blank(),
        axis.text.x = element_text(angle = 15, hjust = 1))
```

Income Status Distribution by Continent (Absolute Frequencies)

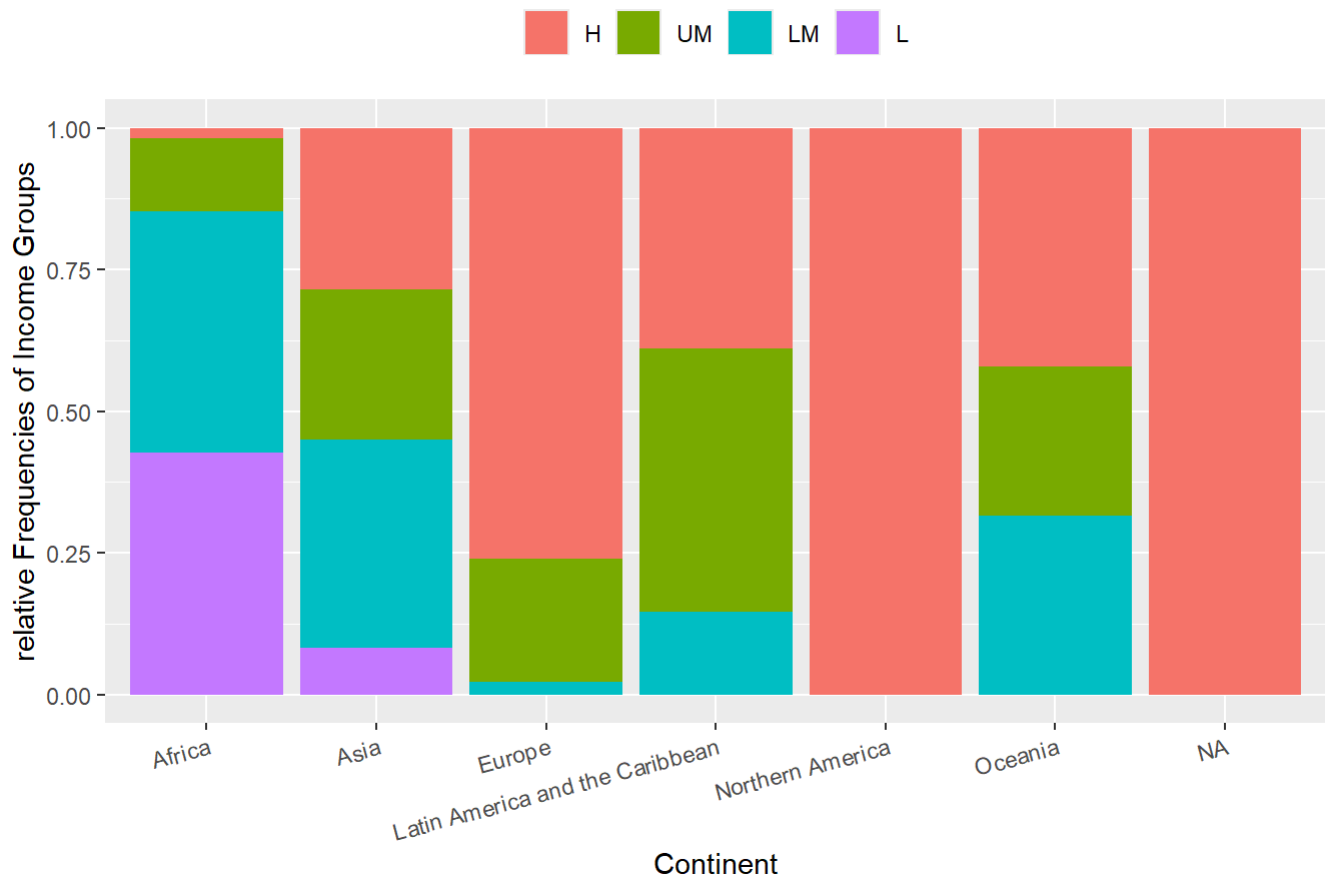


While the absolute barplot shows the actual number of entities per continent, it can be misleading for comparisons. For example, Oceania appears to have very few high-income countries, but proportionally it has more than Asia.

- Create another stacked barplot of relative frequencies (height of the bars should be one). Comment the plot.

```
ggplot(ds, aes(x = continent, fill = income_class)) +
  geom_bar(position = "fill") +
  labs(x = "Continent", y = "relative Frequencies of Income Groups", title = "Income Status Distribution by Continent (Relative Frequencies)") +
  theme(legend.position = "top",
        legend.justification = "center",
        legend.title = element_blank(),
        axis.text.x = element_text(angle = 15, hjust = 1))
```

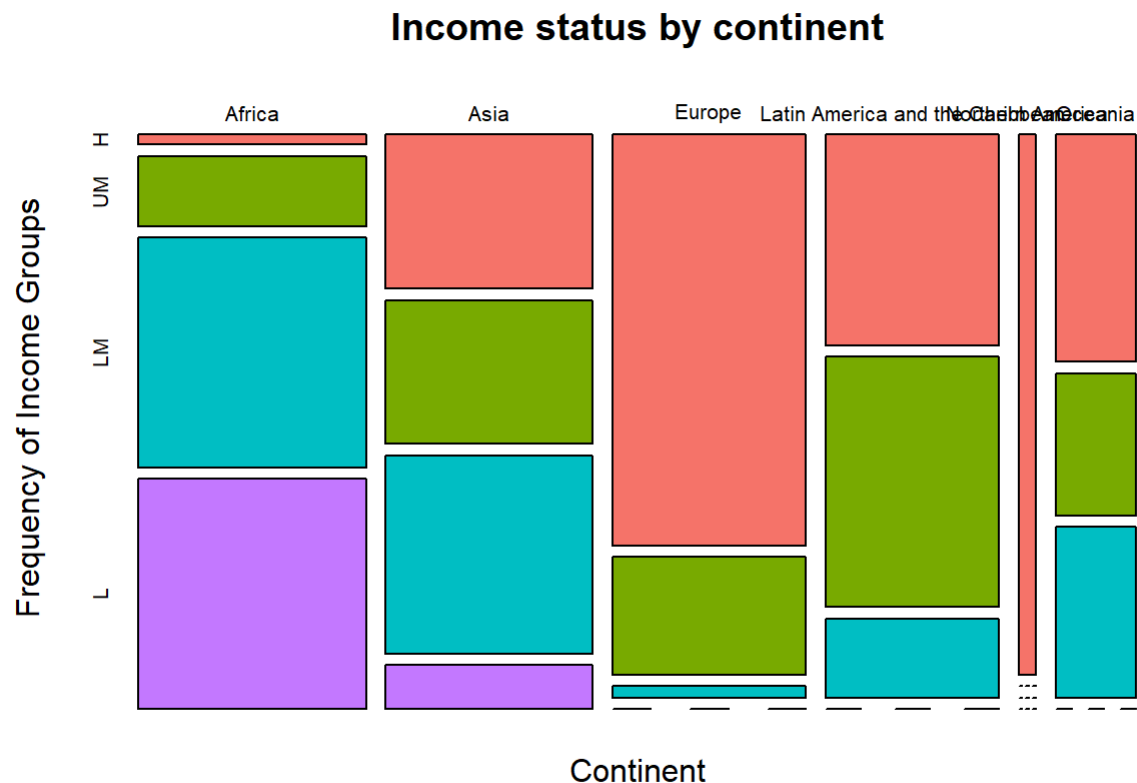
Income Status Distribution by Continent (Relative Frequencies)



Does not take into account how many countries are actually in one income group. Asia for example has more high income countries than North America which is not reflected in this plot.

- Create a mosaic plot of continents and income status using base R functions.

```
mosaicplot(table(ds$continent, ds$income_class),
  main = "Income status by continent",
  xlab = "Continent",
  ylab = "Frequency of Income Groups",
  color = c('#F8766D', '#7CAE00', '#00BFC4', '#C77CFF' ),#same colors as in the above gg2 plots
)
```



- Briefly comment on the differences between the three plots generated to investigate the income distribution among the different continents.

Stacked bar plot of relative frequencies and mosaic are both normalized and show the relative amount of countries in each income group on the y-axis. The mosaic plot also takes into account the number of countries by scaling the width of the bars accordingly. This leads to some unwanted results - x-axis labels overlap which is not easy to correct and the bar for North America looks strange, since it's so thin and only consists of one income group. The stacked bar plot of absolute frequencies takes the number of countries into account by displaying them on the x-Axis.

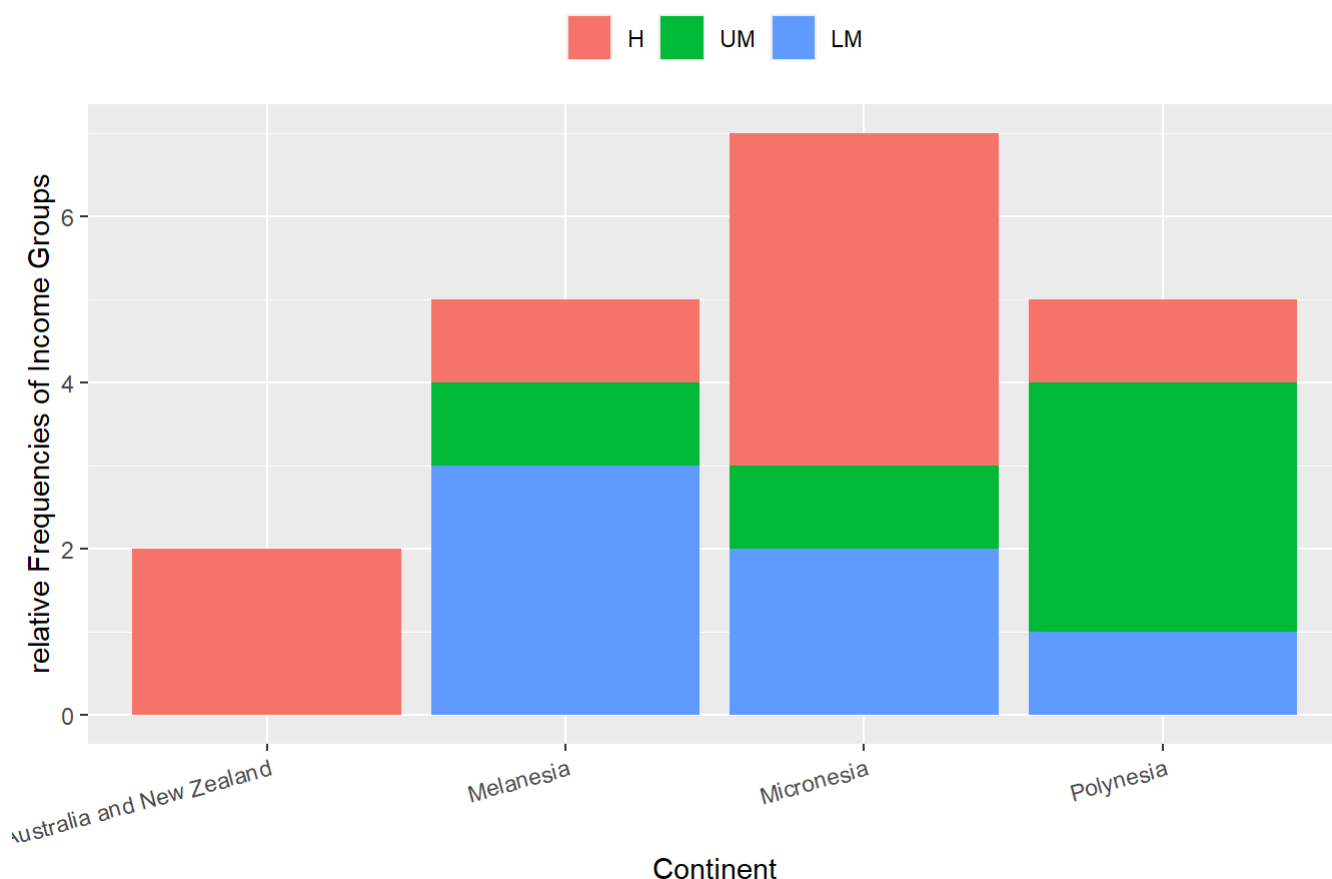
c. Income status in different subcontinents

For Oceania, investigate further how the income status distribution is in the different subcontinents. Use one of the plots in b. for this purpose. Comment on the results.

```
ds_oceania <- ds %>%
  filter(continent == "Oceania")

ggplot(ds_oceania, aes(x = subcontinent, fill = income_class)) +
  geom_bar(position = "stack") +
  labs(x = "Continent", y = "relative Frequencies of Income Groups", title = "Income Status Distribution across Oceania's Subregions") +
  theme(legend.position = "top",
        legend.justification = "center",
        legend.title = element_blank(),
        axis.text.x = element_text(angle = 15, hjust = 1))
```

Income Status Distribution across Oceania's Subregions

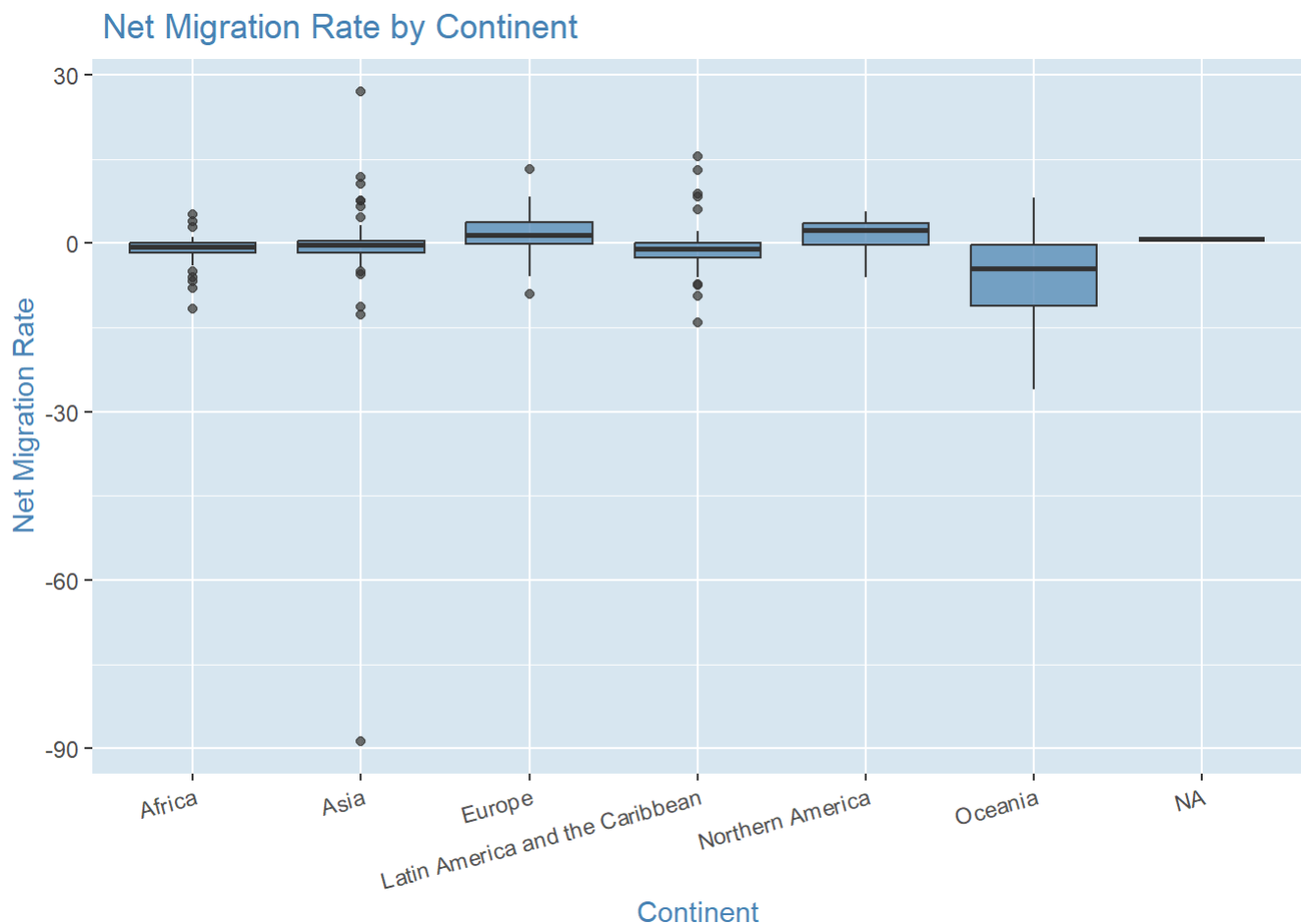


Australia and New Zealand are both high income countries. Melanesia consists of mostly lower-middle income countries. Micronesia consists of roughly half lower-middle and upper-middle income countries; it is also the subcontinent with the most countries. Polynesia consists of mostly upper-middle income countries.

d. Net migration in different continents

- Using **ggplot2**, create parallel boxplots showing the distribution of the net migration rate in the different continents.
- Prettify the plot (change y-, x-axis labels, etc).
- Identify which country in Asia constitutes the largest negative outlier and which country in Asia constitutes the largest positive outlier.
- Comment on the plot.

```
ggplot(ds, aes(x = continent, y = net_migration_rate)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  labs(x = "Continent",
       y = "Net Migration Rate",
       title = "Net Migration Rate by Continent") +
  theme(axis.text.x = element_text(angle = 15, hjust = 1),
        text = element_text(color = "steelblue"),
        panel.background = element_rect(fill = adjustcolor("steelblue", alpha.f = 0.2)))
```



```
# outlier
ds %>%
  filter(continent == "Asia") %>%
  slice_min(net_migration_rate, n = 1) %>%
  select(country, net_migration_rate)
```

country	net_migration_rate
<chr>	<dbl>
Lebanon	-88.7

1 row

```
ds %>%
  filter(continent == "Asia") %>%
  slice_max(net_migration_rate, n = 1) %>%
  select(country, net_migration_rate)
```

country	net_migration_rate
<chr>	<dbl>
Syria	27.1

1 row

Across all continents, the net migration rate for most countries is close to zero. Oceania stands out as the continent with the greatest variability in migration rates. Lebanon represents the most extreme negative outlier with a net migration rate of -89. Syria, on the other hand, constitutes the largest positive outlier with a rate of 27

per 1,000 persons. Latin America and the Caribbean also has a few outliers in the range -15 to 15.

e. Net migration in different subcontinents

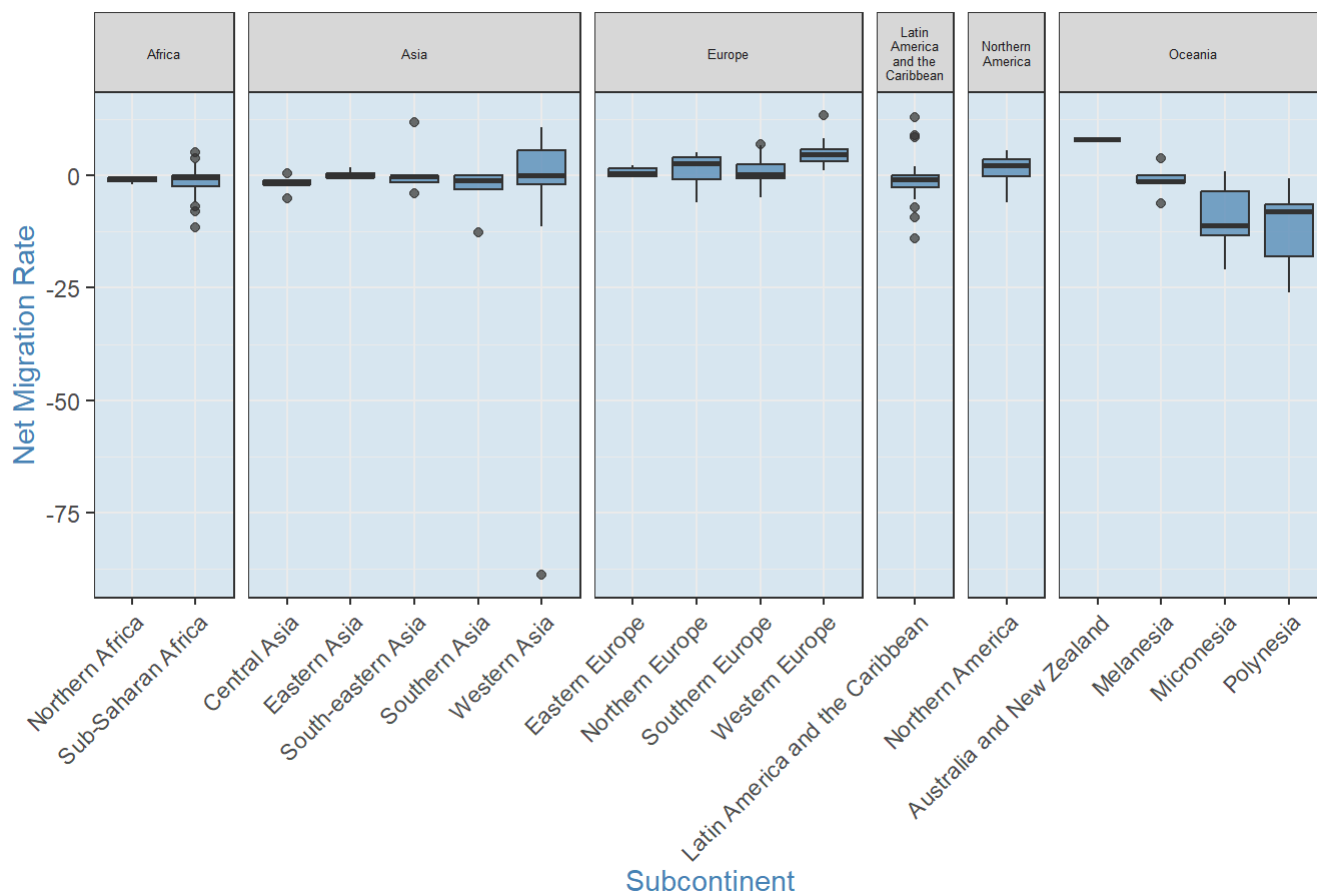
The graph in d. clearly does not convey the whole picture. It would be interesting also to look at the subcontinents, as it is likely that a lot of migration flows happen within the continent.

- Investigate the net migration in different subcontinents using again parallel boxplots. Group the boxplots by continent (hint: use `facet_grid` with `scales = "free_x"`).
- Remember to prettify the plot (rotate axis labels if needed).
- Describe what you see.

```
# remove NA values to prettify
ds_clean <- ds %>% filter(!is.na(subcontinent))

ggplot(ds_clean, aes(x = subcontinent, y = net_migration_rate)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  facet_grid(~ continent, scales = "free_x", space = "free_x", labeller = label_wrap_gen(widht
h = 10)) +
  labs(x = "Subcontinent",
       y = "Net Migration Rate",
       title = "Net Migration Rate by Subcontinent") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 5),
        text = element_text(color = "steelblue"),
        panel.background = element_rect(fill = adjustcolor("steelblue", alpha.f = 0.2)))
```

Net Migration Rate by Subcontinent



das gehört auch noch zu spread out21:34 Claude hat geantwortet: > Lebanon is again such an extreme outlier that it makes all other values comparatively small. Lebanon is again such an extreme outlier that it makes all other values comparatively small. Net migration rates in Western Asia, as well as in Micronesia and Polynesia, are in general a lot more spread out than in other subregions. Micronesia and Polynesia tend to be generally negative, however these regions consist of micronations which often skew statistics. Western Europe and Australia and New Zealand have the highest positive net migration rates. Latin America and the Caribbean stands out as it has many countries whose migration rates are outliers, in both positive and negative directions.

f. Median net migration rate per subcontinent.

The plot in task e. shows the distribution of the net migration rate for each subcontinent. Here you will work on visualizing only one summary statistic, namely the median.

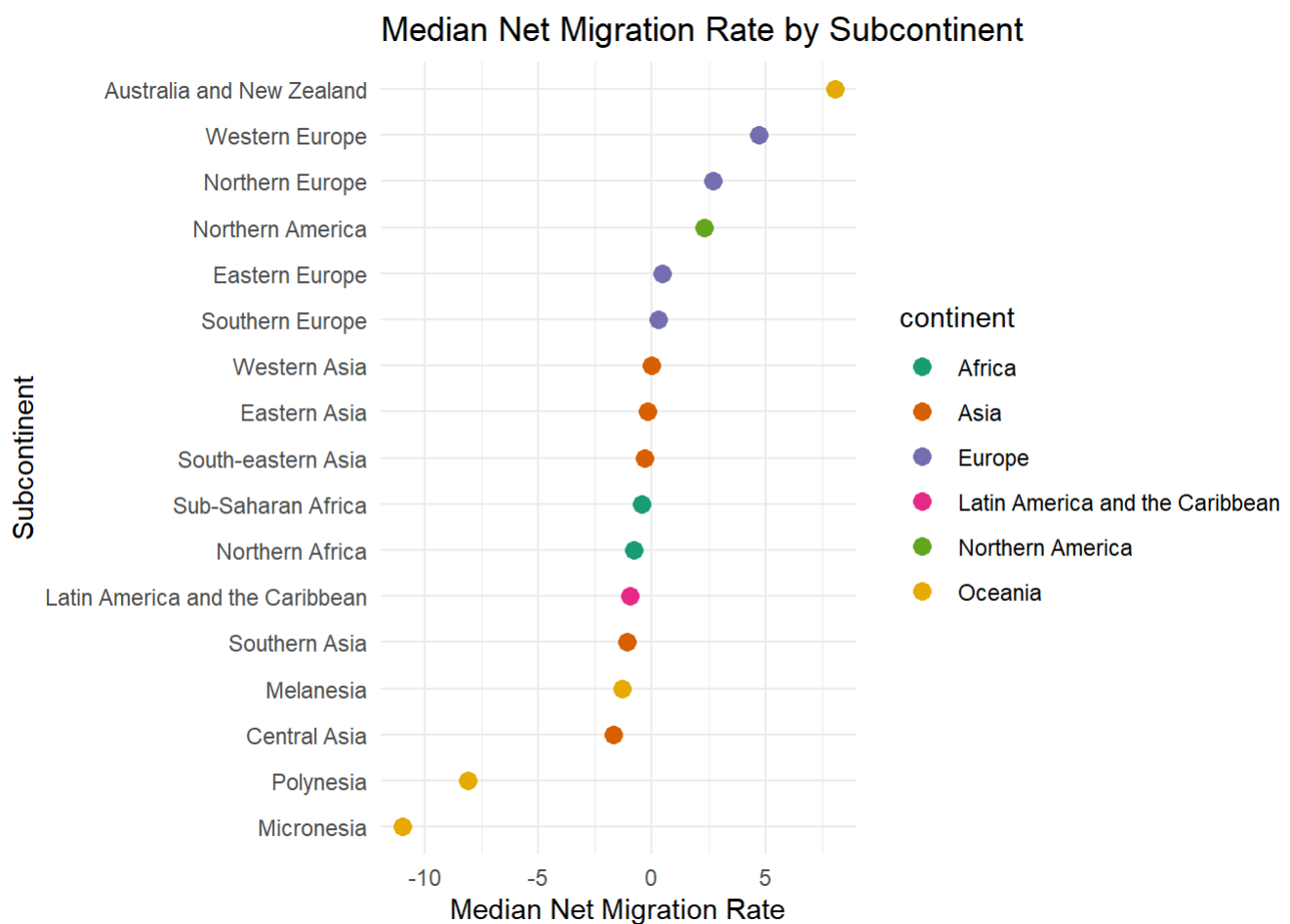
For each subcontinent, calculate the median net migration rate. Then create a plot which contains the subregions on the y-axis and the median net migration rate on the x-axis.

- As geoms use points.
- Color the points by continent – use a colorblind friendly palette (see e.g., here ([http://www.cookbook-r.com/Graphs/Colors_\(ggplot2\)/](http://www.cookbook-r.com/Graphs/Colors_(ggplot2)/))).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median net migration rate and the upper most region contains the highest median net migration rate.
- Comment on the plot. E.g., what are the regions with the most influx? What are the regions with the most outflux?

```
median_migration <- ds %>%
  filter(!is.na(subcontinent)) %>%
  group_by(continent, subcontinent) %>%
  summarise(
    median_migration = median(net_migration_rate, na.rm = TRUE)
  )
```

```
## `summarise()` has grouped output by 'continent'. You can override using the
## `.groups` argument.
```

```
ggplot(
  median_migration,
  aes(
    x = median_migration,
    y = fct_reorder(subcontinent, median_migration),
    color = continent
  )
) +
  geom_point(size = 3) +
  scale_color_brewer(palette = "Dark2") +
  labs(
    title = "Median Net Migration Rate by Subcontinent",
    x = "Median Net Migration Rate",
    y = "Subcontinent"
  ) +
  theme_minimal()
```



Australia and New Zealand as well as Western Europe show the highest positive median net migration rates, indicating strong immigration inflows. In contrast, Micronesia and Polynesia show the most negative migration balances. Most subcontinents cluster around values close to zero.

g. Median youth unemployment rate per subcontinent

For each subcontinent, calculate the median youth unemployment rate. Then create a plot which contains the sub-regions on the y-axis and the median unemployment rate on the x-axis.

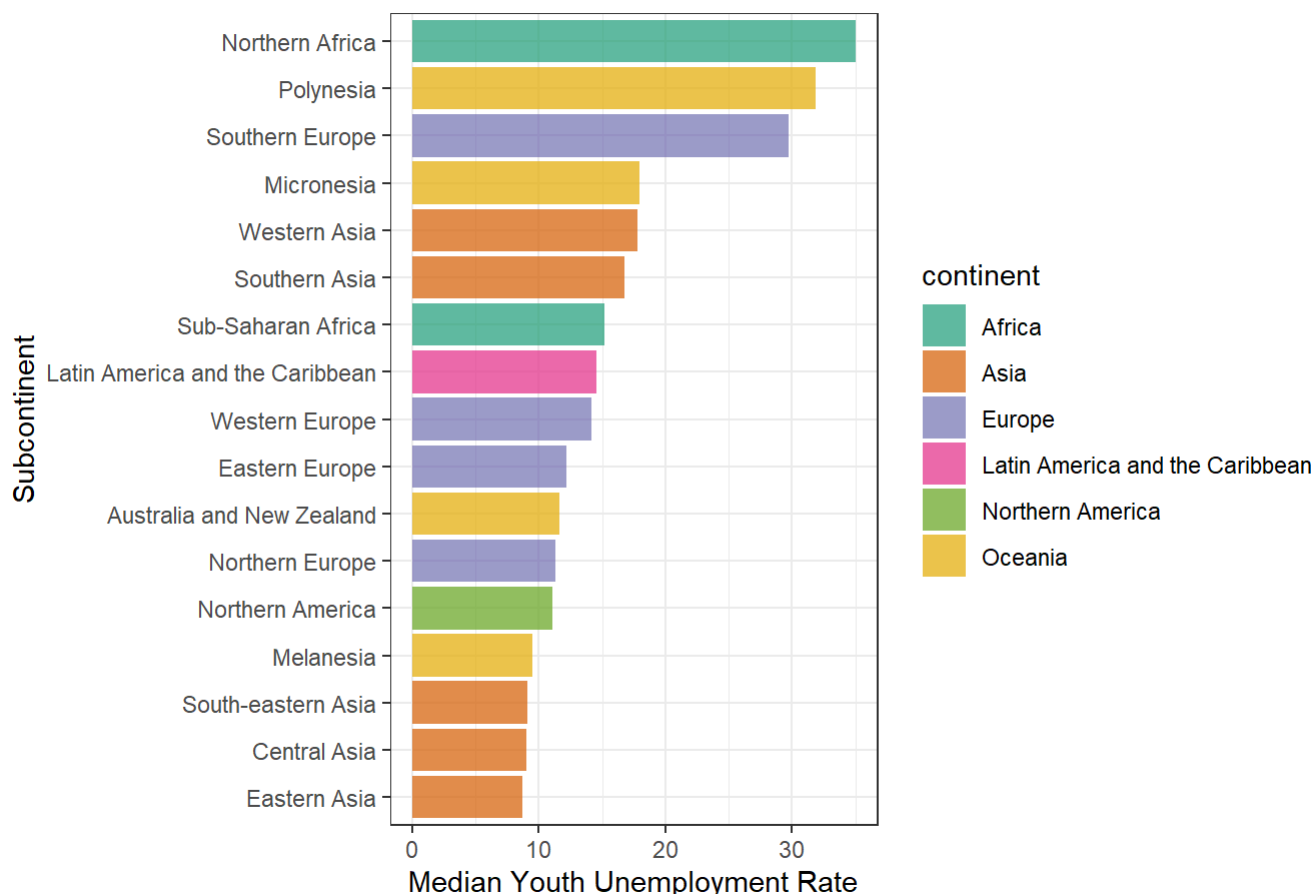
- Use a black and white theme (`?theme_bw()`)

- As geoms use bars. (hint: pay attention to the statistical transformation taking place in `geom_bar()` – look into argument `stat="identity"`)
- Color the bars by continent – use a colorblind friendly palette.
- Make the bars transparent (use `alpha = 0.7`).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median youth unemployment rate and the upper most region contains the highest median youth unemployment rate.
- Comment on the plot. E.g., what are the regions with the highest vs lowest youth unemployment rate?

```
median_unemployment <- ds %>%
  filter(!is.na(subcontinent)) %>%
  group_by(continent, subcontinent) %>%
  summarise(
    median_unemployment = median(youth_unempl_rate, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(
  median_unemployment,
  aes(
    x = median_unemployment,
    y = fct_reorder(subcontinent, median_unemployment),
    fill = continent
  )
) +
  geom_bar(stat = "identity", alpha = 0.7) +
  scale_fill_brewer(palette = "Dark2") +
  labs(
    title = "Median Youth Unemployment Rate by Subcontinent",
    x = "Median Youth Unemployment Rate",
    y = "Subcontinent"
  ) +
  theme_bw()
```

Median Youth Unemployment Rate by Subcontinent



Northern Africa and Polynesia show the highest median youth unemployment rates, whereas Eastern and Central Asia have the lowest values. European subcontinents generally exhibit lower unemployment rates compared to Africa and Oceania.

h. Median youth unemployment rate per subcontinent – with error bars

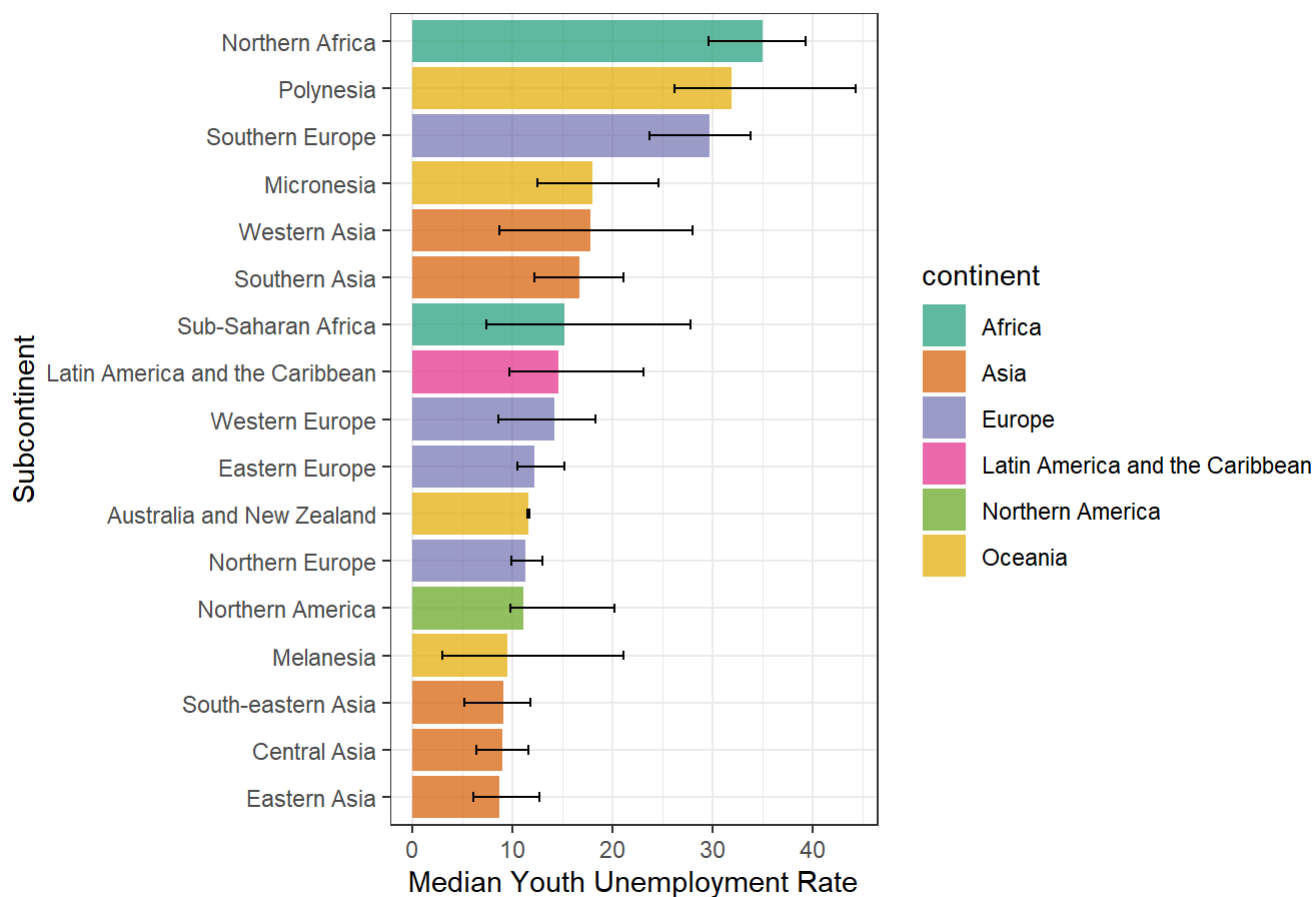
The value displayed in the barplot in g. is the result of an aggregation, so it might be useful to also plot error bars, to have a general idea on how precise the median unemployment is. This can be achieved by plotting the error bars which reflect the standard deviation or the interquartile range of the variable in each of the subcontinents.

Repeat the plot from Task g. but include also error bars which reflect the 25% and 75% quantiles. You can use `geom_errorbar` in **ggplot2**.

```
unemployment_summary <- ds %>%
  filter(!is.na(subcontinent)) %>%
  group_by(continent, subcontinent) %>%
  summarise(
    median_unemployment = median(youth_unempl_rate, na.rm = TRUE),
    q25 = quantile(youth_unempl_rate, 0.25, na.rm = TRUE),
    q75 = quantile(youth_unempl_rate, 0.75, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(
  unemployment_summary,
  aes(
    x = median_unemployment,
    y = fct_reorder(subcontinent, median_unemployment),
    fill = continent
  )
) +
  geom_bar(stat = "identity", alpha = 0.7) +
  geom_errorbar(
    aes(xmin = q25, xmax = q75),
    width = 0.2
  ) +
  scale_fill_brewer(palette = "Dark2") +
  labs(
    title = "Median Youth Unemployment Rate with Interquartile Range",
    x = "Median Youth Unemployment Rate",
    y = "Subcontinent"
  ) +
  theme_bw()
```

Median Youth Unemployment Rate with Interquartile Range



The error bars indicate that the variability of youth unemployment differs substantially across subcontinents. Wider ranges suggest stronger differences between countries within the same region.

i. Relationship between education expenditure and net migration rate

Using **ggplot2**, create a plot showing the relationship between education expenditure and net migration rate.

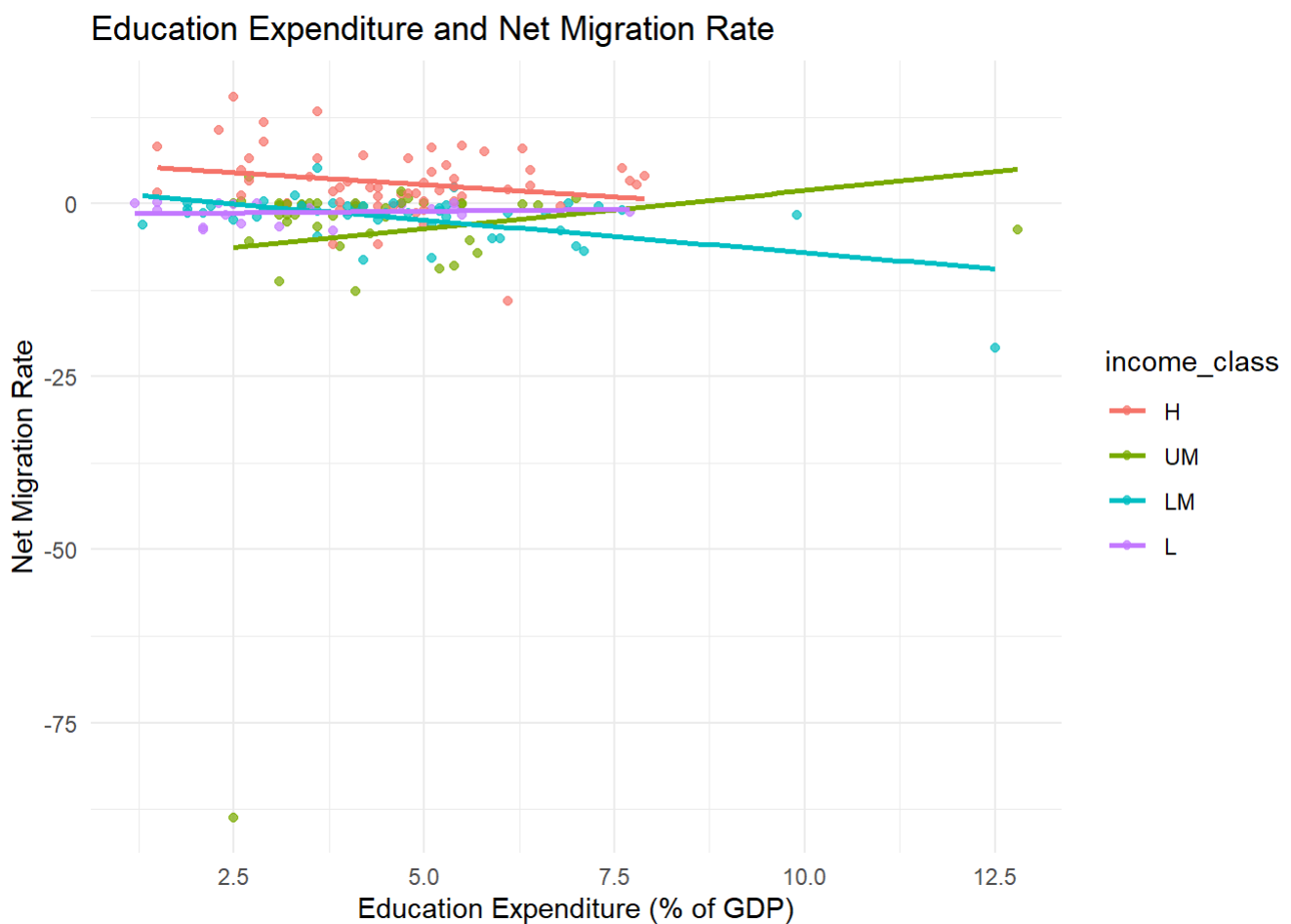
- Color the geoms based on the income status.
- Add a regression line for each development status (using `geom_smooth()`).

Comment on the plot. Do you see any relationship between the two variables? Do you see any difference among the income levels?

```
ds_edu <- ds %>%
  filter(
    !is.na(education_expenditures),
    !is.na(net_migration_rate)
  )

ggplot(
  ds_edu,
  aes(
    x = education_expenditures,
    y = net_migration_rate,
    color = income_class
  )
) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Education Expenditure and Net Migration Rate",
    x = "Education Expenditure (% of GDP)",
    y = "Net Migration Rate"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The relationship between education expenditure and net migration appears relatively weak overall. High-income countries tend to combine higher education expenditures with more positive migration balances.

j. Relationship between youth unemployment and net migration rate

Create a plot as in Task i. but for youth unemployment and net migration rate. Comment briefly.

```
ds_unempl <- ds %>%
  filter(
    !is.na(youth_unempl_rate),
    !is.na(net_migration_rate)
  )

ggplot(
  ds_unempl,
  aes(
    x = youth_unempl_rate,
    y = net_migration_rate,
    color = income_class
  )
) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Youth Unemployment Rate and Net Migration Rate",
    x = "Youth Unemployment Rate",
    y = "Net Migration Rate"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Youth Unemployment Rate and Net Migration Rate



Countries with higher youth unemployment rates tend to lead to more negative migration balances. The overall relationship between the two variables appears relatively weak.

k. Merging population data

Go online and find a data set which contains the 2020 population for the countries of the world together with ISO codes.

- Download this data and merge it to the dataset you are working on in this case study using a left join. (A possible source: World Bank (<https://data.worldbank.org/indicator/SP.POP.TOTL?end=2020&start=2020>)))
- Inspect the data and check whether the join worked well.

```
population <- read.csv('Pop_worldbank.csv', skip = 4) %>%
  select(Country.Code, 'population' = X2020)
head(population)
```

	Country.Code <chr>	population <dbl>
1	ABW	108587
2	AFE	694446100
3	AFG	39068979
4	AFW	474569351
5	AGO	33451132

	Country.Code <chr>	population <dbl>
6	ALB	2528480

6 rows

```
ds_inkl_pop <- left_join(ds, population, by = c('GENC' = 'Country.Code'), as)
head(ds_inkl_pop)
```

country <chr>	G... <chr>	continent <chr>	education_expenditures <dbl>	net_migration_rate <dbl>	youth_... <dbl>
1 Afghanistan	AFG	Asia	4.1	-0.1	
2 Albania	ALB	Europe	3.6	-3.3	
3 Algeria	DZA	Africa	NA	-0.9	
4 American Samoa	ASM	Oceania	NA	-26.1	
5 Andorra	AND	Europe	3.2	0.0	
6 Angola	AGO	Africa	3.4	-0.2	

6 rows | 1-7 of 10 columns

To check how good the join worked, we filtered the joined dataset for NA values in the population column. This shows, that only for Taiwan, no population data is available.

```
filter(ds_inkl_pop, is.na(ds_inkl_pop$population))
```

country <chr>	G... <chr>	continent <chr>	education_expenditures <dbl>	net_migration_rate <dbl>	youth_unempl_rat <dbl>
Taiwan	TWN	NA	NA	0.8	NA

1 row | 1-7 of 9 columns

I. Scatterplot of education expenditure and net migration rate in Europe

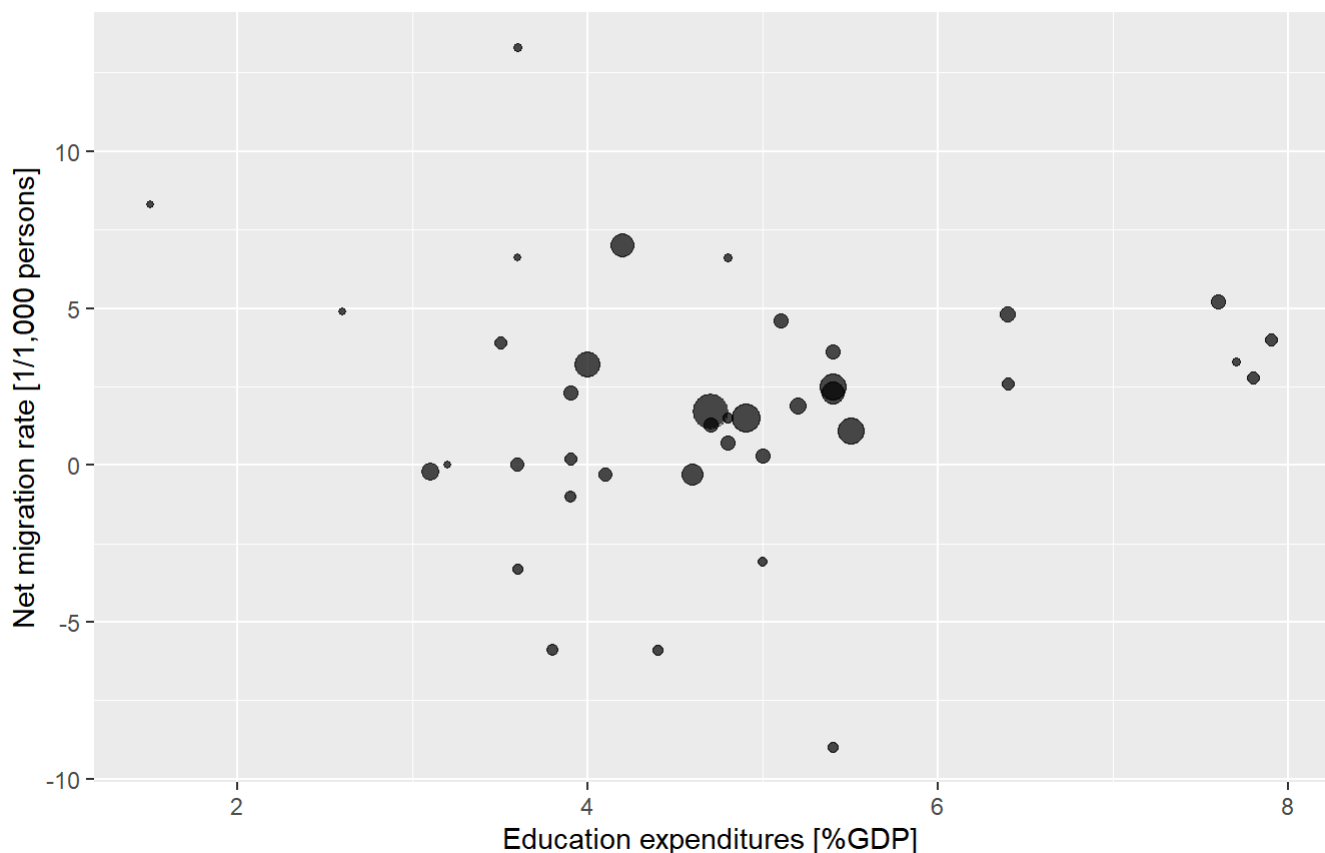
Make a scatterplot of education expenditure and net migration rate for the countries of Europe.

- Scale the size of the points according to each country's population.
- For better visibility, use a transparency of `alpha=0.7`.
- Remove the legend.
- Comment on the plot.

```
ds_inkl_pop %>%
  filter(continent == 'Europe') %>%
  ggplot(aes(education_expenditures, net_migration_rate, size = population)) +
  geom_point(alpha = 0.7) +
  labs(title = 'Net migration rate vs. Education expenditures', subtitle = 'in European countries scaled by population ', x = 'Education expenditures [%GDP]', y = 'Net migration rate [1/1,000 persons]') +
  theme(legend.position = 'none')
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Net migration rate vs. Education expenditures
in European countries scaled by population



The scatter plot suggests that countries with a higher investment in education also have a higher net migration rate. There is also a cluster of points around a 5% education expenditure and due to their relatively large size, this suggests that a large part of Europe's population is located there.

m. Interactive plot

On the merged data set from Task k., using function `ggplotly` from package **plotly** re-create the scatterplot in Task l., but this time for all countries. Color the points according to their continent.

When hovering over the points the name of the country, the values for education expenditure, net migration rate, and population should be shown. (Hint: use the aesthetic `text = Country`. In `ggplotly` use the argument `tooltip = c("text", "x", "y", "size")`).

```
#install.packages('plotly')
library(plotly)
```

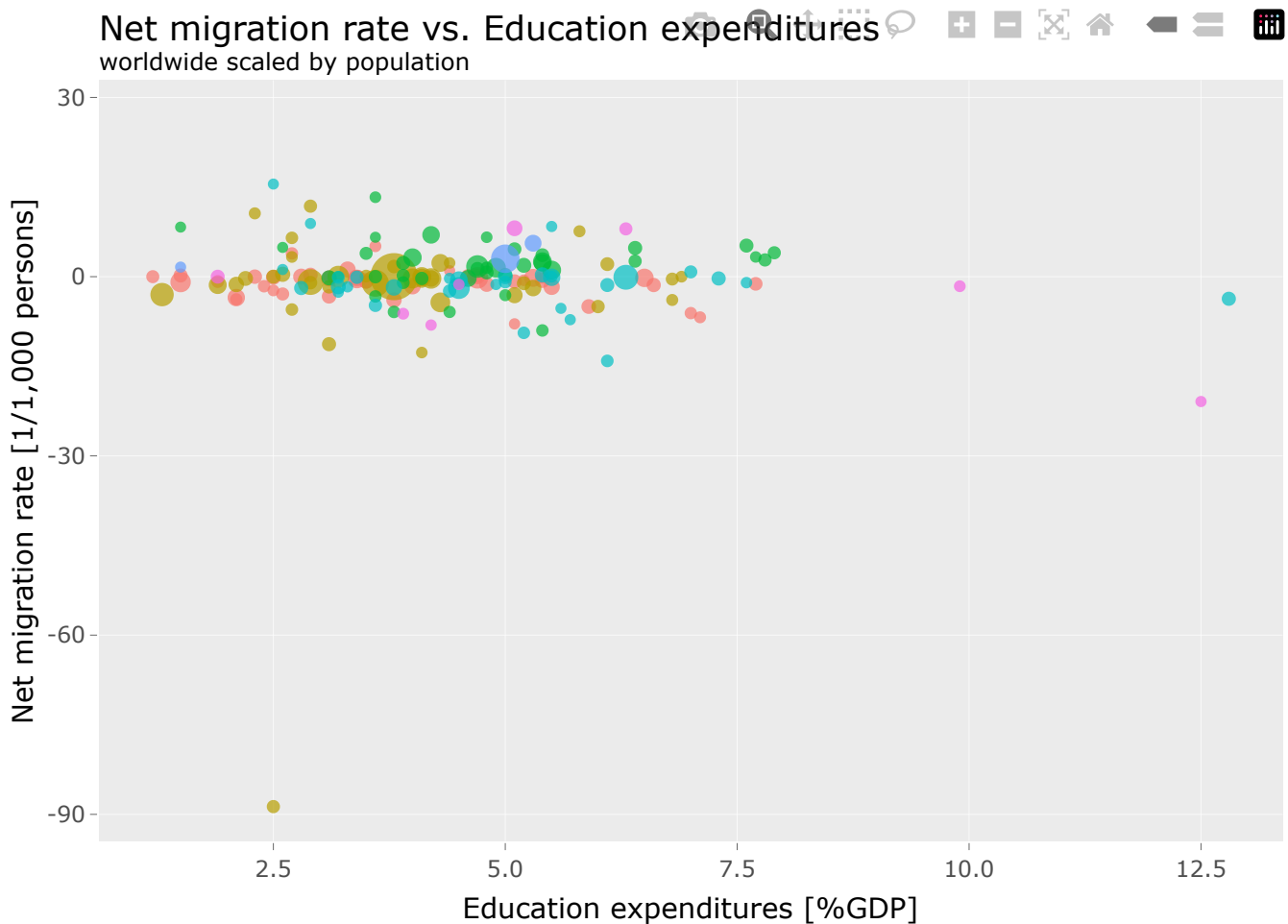
```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##   last_plot
```

```
## The following object is masked from 'package:stats':
##
##   filter
```

```
## The following object is masked from 'package:graphics':
##
##   layout
```

```
scatter_pop <- ds_inkl_pop %>%
  ggplot(aes(education_expenditures, net_migration_rate, size = population, color = continen
t, text = paste('Country: ', country, '<br>Education expenditures: ', education_expenditures,
'<br>Net migration rate: ', net_migration_rate, '<br>population: ', population))) +
  geom_point(alpha = 0.7) +
  theme(legend.position = 'none') +
  labs(title = 'Net migration rate vs. Education expenditures<br><sup>worldwide scaled by pop
ulation', x = 'Education expenditures [%GDP]', y = 'Net migration rate [1/1,000 persons]')
ggplotly(scatter_pop, tooltip = c("text"))
```



The plot shows, that most countries are distributed around the 0 persons net migration rate and spread over a education expenditure of 1,2 - 7,0% GDP. A drastic outlier regarding the net migration rate is Lebanon with -88.7/1000 persons. For the education expenditures the outliers are the Solomon Islands, the Federated States of Micronesia and Cuba with >9% of their GDP.

Hint: If you are creating a .pdf file, keep in mind that PDF output is static and does not support HTML widgets. To avoid build errors, you can wrap interactive plots in a conditional check or set the chunk option `eval=FALSE`. Alternatively, you can use the **webshot** package to automatically convert interactive elements into static images during the knit process.

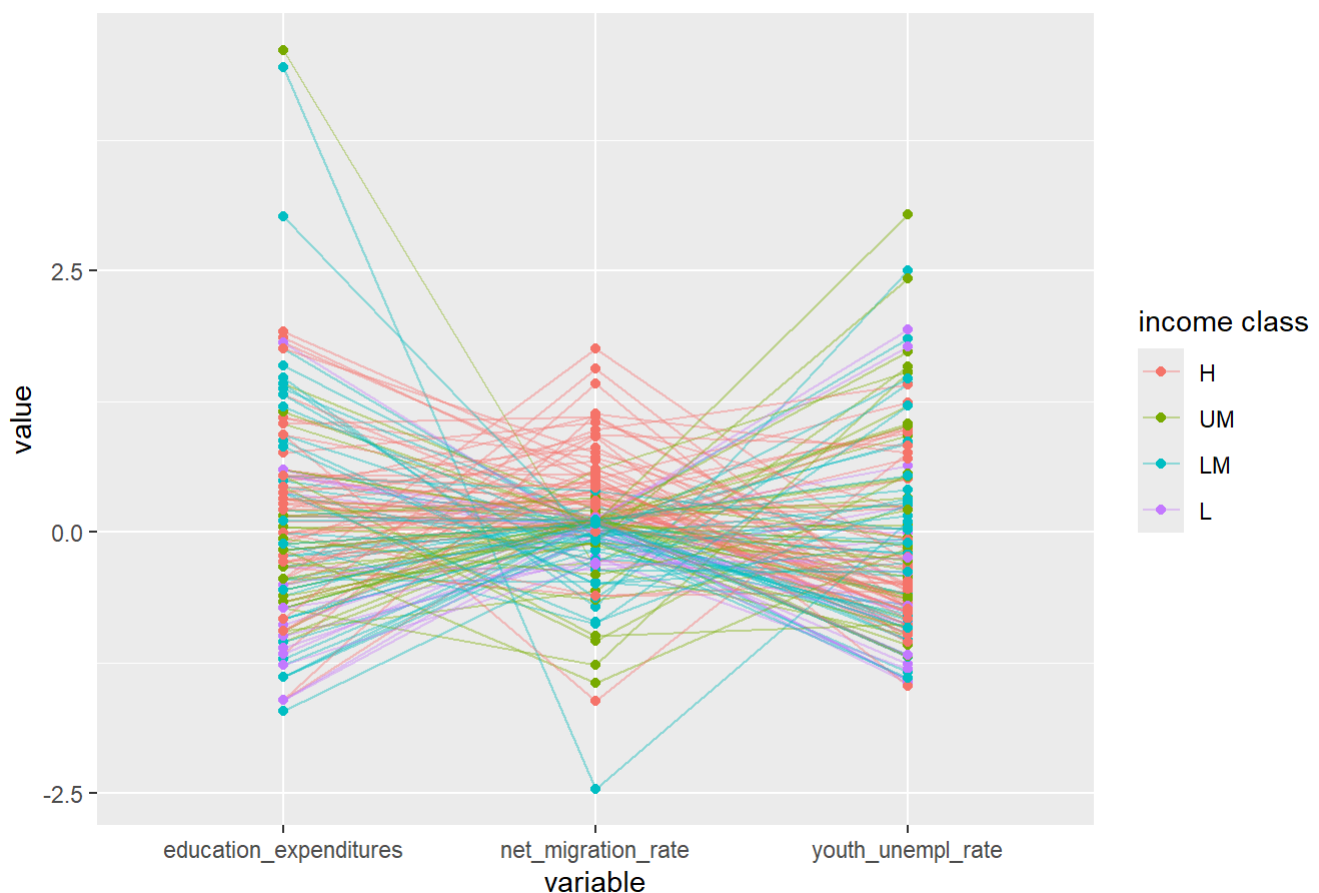
n. Parallel coordinate plot

In **parallel coordinate plots** each observation or data point is depicted as a line traversing a series of parallel axes, corresponding to a specific variable or dimension. It is often used for identifying clusters in the data.

One can create such a plot using the **GGally** R package. You should create such a plot where you look at the three main variables in the data set: education expenditure, youth unemployment rate and net migration rate. Color the lines based on the income status. Briefly comment.

```
#install.packages('GGally')
library('GGally')
ggparcoord(ds, columns = 4:6, groupColumn = 7, showPoints = TRUE, alpha = 0.4, title = 'Comparison of all three variables by income status', scale = 'std')+
  labs(colour = 'income class')
```

Comparison of all three variables by income status



The most obvious cluster in the data is the one of a high net migration rate of high income countries. These also tend to have a high education expenditure and low youth unemployment rate. For the low income

countries a slight cluster around a low education expenditure is shown. It also can be seen that some of the highest education expenditures and lowest net migration rates are presented by upper and lower middle income countries.

o. World map visualisation

Create a world map of the education expenditure per country. You can use the vignette <https://cran.r-project.org/web/packages/rworldmap/vignettes/rworldmap.pdf> (<https://cran.r-project.org/web/packages/rworldmap/vignettes/rworldmap.pdf>) to find how to do this in R. Alternatively, you can use other packages (such as **ggplot2**, **sf** and **rnaturalearthdata**) to create a map.

```
#install.packages('rworldmap')
library(rworldmap)
```

```
## Loading required package: sp
```

```
## ### Welcome to rworldmap ###
```

```
## For a short introduction type :  vignette('rworldmap')
```

```
head(ds)
```

country <chr>	G... continent <chr><chr>	education_expenditures <dbl>	net_migration_rate <dbl>	youth_...
1 Afghanistan	AFG Asia	4.1	-0.1	
2 Albania	ALB Europe	3.6	-3.3	
3 Algeria	DZA Africa	NA	-0.9	
4 American Samoa	ASM Oceania	NA	-26.1	
5 Andorra	AND Europe	3.2	0.0	
6 Angola	AGO Africa	3.4	-0.2	

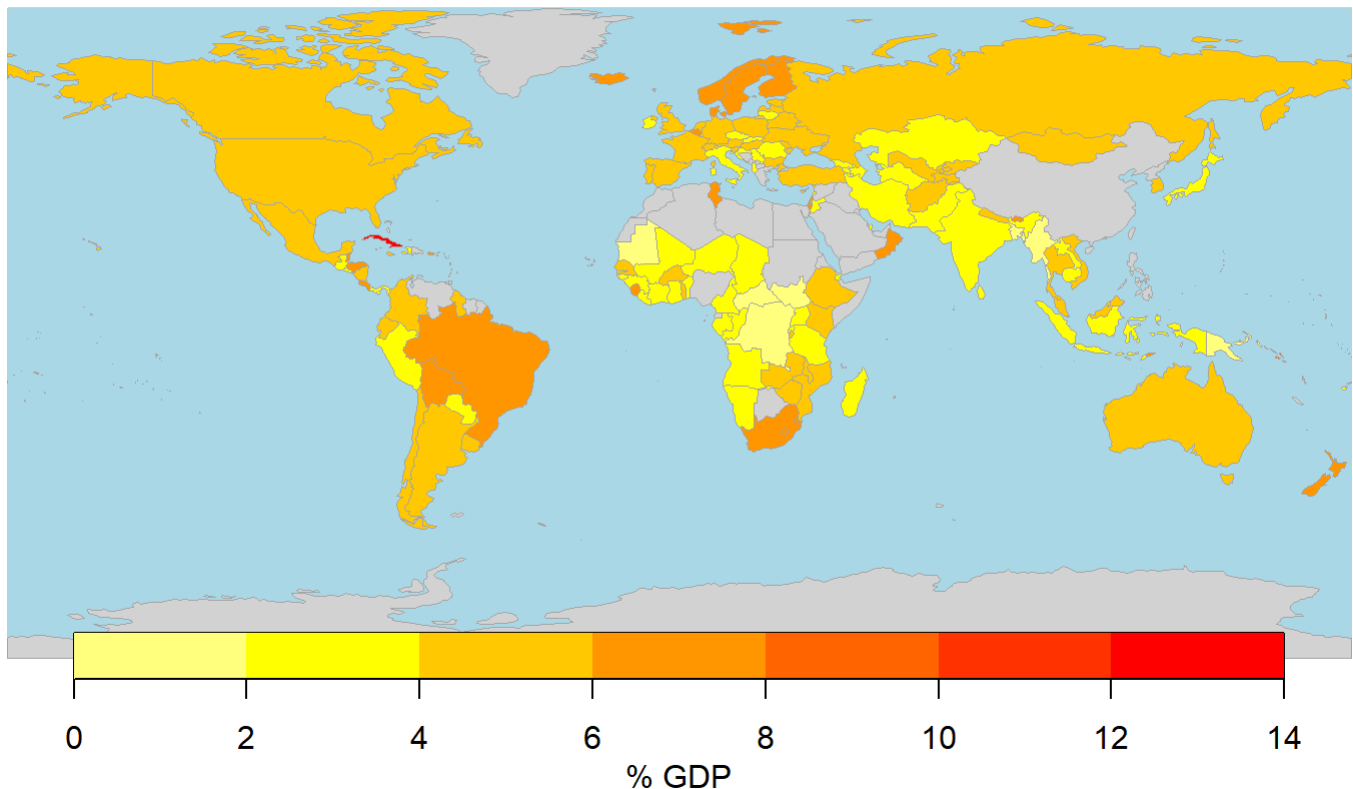
6 rows | 1-7 of 9 columns

```
mapEE <- joinCountryData2Map(ds, nameJoinColumn = 'GENC')
```

```
## 213 codes from your data successfully matched countries in the map
## 1 codes from your data failed to match with a country code in the map
## 30 codes from the map weren't represented in your data
```

```
par(mai = c(.2,0,0.2, 0), xaxs = 'i', yaxs = 'i')
map_params <- mapCountryData(mapEE, nameColumnToPlot = 'education_expenditures', mapTitle =
'Education Expenditures per country', oceanCol = 'lightblue', missingCountryCol = 'light gre
y', borderCol = 'darkgrey', addLegend = FALSE, catMethod = 'pretty')
mtext(text = '% GDP', side = 1, line = -0.5)
do.call(addMapLegend, c(map_params, legendLabels = 'all', legendIntervals = 'data', legendMar
= 3.5))
```

Education Expenditures per country



The map shows that the highest education expenditures can be found in Brazil and Bolivia, South Africa and northern Europe. Cuba also stands out again with its high education expenditure and bright red coloring. The lowest values are distributed over Africa, Central, South and Southeast Asia.